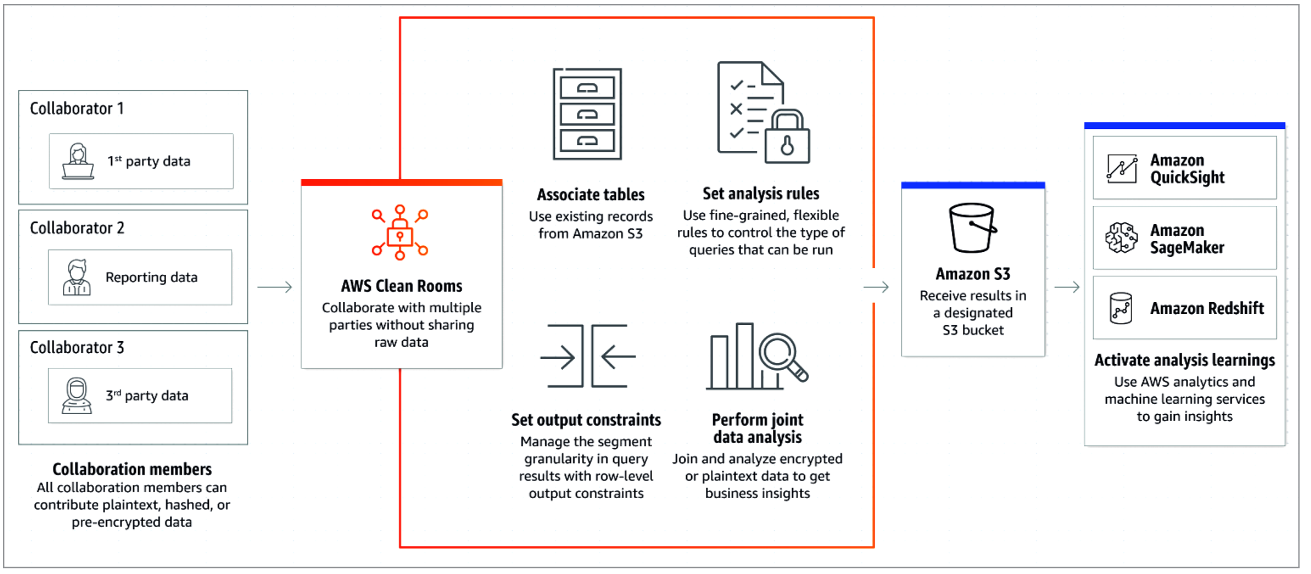

Fig. 2: Amazon Clean Rooms

unified map of data assets and their relationships

- Data discovery to help businesses do smart search
- Strong visibility into sensitive data across the entire data bank
- Strong central management of data

The architecture of Microsoft Purview is composed of:

- Data map
- Data catalogue
- Data estate insights
- Data sharing

AWS Clean Rooms: Data management solution on the AWS platform

With Amazon's AWS Clean Rooms (Fig. 2), we can collaborate with multiple partners and clients to prepare data analytics, reports, and insights. This can be achieved by sharing data with protection, thus maintaining data privacy for all the concerned parties.

We can use AWS Clean Rooms to share data with partners or vice-versa as plain text, hashed data dump on S3 bucket, or pre-encrypted data. It can be configured easily for multiple collaborators without revealing raw data of the partner organisation. We can perform joint analysis with AWS Clean Rooms for research or market analysis, or improved reporting.

This kind of service helps to play around with real customer data for data analytics in a sandbox setup, before the actual live data is experienced in public for data analysis. This helps to improve reporting and monitoring, as well as product development.

We can use templated queries for controlling data operations, flexible rules to handle data and its access management, and associate existing data (e.g., historical data analytics) for collaborative data analytics. We can use AWS machine learning services like SageMaker and Quicksight for deep dive analytics and insights from multiple collaborative data using AWS Clean Rooms.

Data shared by partners and clients in AWS Clean Rooms is fully secure and protected from unauthorised access. It is encrypted, safe to store, and safe to clean and remove from AWS Clean Rooms. It supports file formats like csv and parquet, and can be pre-encrypted before being transported for data analytics.

Since it has been released only recently, it is available only in selected regions; we can expect general availability (GA) of AWS Clean Rooms soon.

Visual data exploration in Amazon Neptune

Graph databases are popular for both transactional data (OLTP) and analytical data (OLAP) visualisation, for a quick analysis of entity relations and data dependencies. Amazon Neptune (Fig. 3) is a popular, fully-managed graph database that is fast and reliable for interpreting data visually. It helps to visually interpret highly connected data sets, prepare knowledge graphs, and navigate data easily.

Amazon Neptune is a low-latency graph database with multi-regional and multi-availability zone replication, continuous backup to Amazon S3 bucket for file streaming and data feeds, and point-in-time recovery and read replicas for high throughput with queries. It uses the Apache TinkerPop framework for graph visualisation of data set and graph analytic capabilities, and uses Apache TinkerPop Gremlin as a graph traversal language to express